

Name _____
Physics

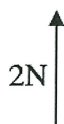
Date _____

~~Mrs. Rush~~

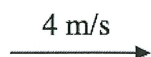
Finding components:

1. The most useful components in which to break a single vector into are _____.
2. A given vector can be broken into how many components.
3. A force of 40 N is acting at 30° . Draw this vector and find its horizontal and vertical components.
4. You are driving 29 m/s (about 65 mph) in a direction of 60° North of West (120°). Draw this vector and find both components.
5. You walk 10 meters (at 315° degrees) or in a direction of 45° South of East. How far South have you walked, and how far East? [Hint: Draw the vector, and resolve into its horizontal and vertical components!]
6. Find the x and y components (horizontal and vertical components) of each of the following: ****not drawn to scale****

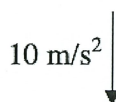
a.)



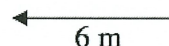
b.)



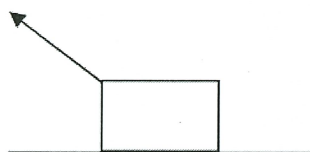
c.)



d.)



7. A block is pulled along a horizontal surface as show below. If the resultant force is 12 newtons at an angle of 40° from the horizontal, find the force moving the block forward and the force pulling the block upward. Solve both graphically and algebraically.



Adding Vectors

8. Consider the following:

- a.) Andrew combines 3 and 4 and gets 7.
- b.) Alexa combines 3 and 4 and gets 1.
- c.) Alexandra combines 3 and 4 and gets 8.
- d.) Sam combines 3 and 4 and gets 0.
- e.) Paulo combines 3 and 4 and gets 5.

Are they all correct? Why or Why not? How is that possible?

Vector Addition Worksheet

On a separate piece of paper, use the following individual vectors to GRAPHICALLY find the resultant vector in the first three problems. Remember, the resultant vector must have both magnitude and direction. Include a scale.

$$\mathbf{A} = 35 \text{ km at } 25^\circ \text{ N of E}$$

$$\mathbf{C} = 20 \text{ km at } 43^\circ \text{ S of E}$$

$$\mathbf{B} = 15 \text{ km at } 10^\circ \text{ E of N}$$

$$\mathbf{D} = 40 \text{ km at } 28^\circ \text{ S of W}$$

1. $\mathbf{A} + \mathbf{B}$
2. $\mathbf{B} + \mathbf{C}$
3. $\mathbf{C} + \mathbf{D}$

Solve all the following problems MATHEMATICALLY using your calculator. Show all work. Remember, the resultant vector must have both magnitude and direction (an angle).

4. Now, let's assume that **A, B, C, and D** from above are each the resultant of 2 component vectors (x and y). Start by drawing each of these 4 vectors separately and then determine what 2 vectors would be needed to produce each one of them.
5. A boat is heading across a river at a **velocity** of 25 mph. The river is flowing downstream at 10 mph. What is the actual **velocity** of the boat?
6. An airplane is flying 340 km/hr at 12° East of North. The wind is blowing 40 km/hr at 34° South of East. What is the plane's actual **velocity**?
7. You are on an elevator that is plummeting toward the ground at 30 m/s and you jump up right before it hits the ground. You jump upward with a **velocity** of 5 m/s. At what speed do you hit the ground?
8. You push on a box with a **force** of 500 Newtons directly north. Another person pushes the box with a **force** of 800 Newtons directly east. What is the **resultant** force?